# Cyclistic Bike-Share Case Study

How Does a Bike-Share Navigate Speedy Success?

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## Business Problem

Cyclistic aims to increase the number of annual memberships, as annual members are significantly more profitable than casual riders.

The business task is to analyze behavioral differences between annual members and casual riders using historical trip data in order to identify patterns that can inform targeted marketing strategies designed to convert casual riders into annual members.

The key stakeholders for this analysis are the Director of Marketing and the Executive Team, who will use the findings to approve and implement future marketing initiatives.

## Data Source and Preparation

The data used in this analysis consists of 12 months of historical trip data from the Divvy bike-share system. The datasets are publicly available and were made available by Motivate International Inc. for analysis purposes through the Divvy bike-share program.

The datasets contain trip-level information such as ride start and end times, station names, and rider type (member or casual). The data is organized in monthly CSV files, with each file representing trips recorded during a specific month.

No personally identifiable information (PII) is included in the dataset, ensuring compliance with privacy regulations.

The credibility of the dataset was evaluated using the ROCCC framework (Reliable, Original, Comprehensive, Current, and Cited).

These datasets are appropriate for analyzing behavioral differences between annual members and casual riders.

# Tools Used

This project was carried out using SQLite in DBeaver to query, extract, and process a large dataset of more than 5.6 million records. Microsoft Excel was used to support intermediate data exploration, summarize results, group variables, and create the visualizations used in the report.

# Data Cleaning and Processing

## Data Preparation and Integration

The dataset consisted of 12 separate monthly CSV files representing the year 2025. Each file contained trip-level records from the Divvy bike-share system with an identical structure.

The following steps were performed:

1. All monthly CSV files were imported into SQLite using DBeaver.
2. Each file was initially stored as a separate table.
3. All monthly datasets were combined into a single unified table named **trips** using SQL (UNION ALL).

This resulted in a consolidated dataset containing approximately **5.6 million ride records**, which served as the base for further validation and cleaning.

## Data Quality Checks

Before performing any transformations, several validation checks were conducted on the raw dataset to ensure data integrity.

Duplicate Records

The uniqueness of the ride_id column was verified.
No duplicate records were detected.

Missing Values

Key columns were checked for missing values, including:

- ride_id

- started_at
- ended_at
- member_casual

No missing values were found in these critical fields.

Categorical Validation

The following columns were validated to ensure consistency:

- User Type (member_casual)
  Only two valid categories were found:
    - member
    - casual
- Bike Type (rideable_type)
  Only valid bike types were detected:
    - classic_bike
    - electric_bike

This confirms that the dataset is consistent with the expected system configuration.

## Data Issues and Corrections

During the validation process, a data inconsistency was identified in the timestamp fields.

Although the dataset was labeled as part of 2025, a subset of records contained timestamps corresponding to January 2026. A detailed inspection revealed that these records covered the full month (January 1–31), indicating a year-labeling issue rather than additional data.

To ensure consistency with the analysis period (January–December 2025), the timestamps were corrected by subtracting one year from both:

- started_at
- ended_at

The dataset was then revalidated to confirm proper alignment within the 2025 timeframe.

## Feature Engineering

After resolving timestamp inconsistencies, new variables were created to support behavioral analysis:

- Ride Length (ride_length)
  Calculated as the duration of each trip in minutes using the difference between start and end timestamps.
- Day of Week (day_of_week)
  Extracted from the ride start timestamp.
- Ride Hour (ride_hour)
  Represents the starting hour of each ride.
- Ride Month (ride_month)
  Used to analyze seasonal trends.

All derived variables were calculated **after correcting timestamps** to ensure full consistency and accuracy.

## Outlier and Duration Analysis

Following feature engineering, ride durations were evaluated to detect anomalies:

- 29 rides were found to have zero or negative duration and were removed, as they represent invalid records.
- Approximately 5,700 rides exceeded 24 hours, with none exceeding 27 hours.

These longer rides were retained, as they may represent valid extended usage and constitute a very small proportion of the dataset.

## Data Standardization

To ensure accurate calculations and proper aggregation, data types were standardized across relevant columns:

- ride_length was stored as a numeric value
- Time-based variables (day_of_week, ride_hour, ride_month) were stored as integers
- Geographic coordinates were converted to numeric format

- Categorical fields were preserved as text

This standardization ensures reliable filtering, grouping, and analytical operations.

## Final Clean Dataset

A final cleaned dataset named trips_cleaned was created, containing:

- All valid ride records
- Corrected timestamps
- Engineered features (ride_length, day_of_week, ride_hour, ride_month)
- Standardized data types

The dataset was validated to ensure:

- Correct value ranges
- Consistent formatting
- Absence of invalid or corrupted records

It is now fully prepared for the Analyze phase.

# Data Analysis

## Analytical Approach

The analysis was conducted in a structured, step-by-step manner, starting with high-level metrics and then moving into deeper behavioral insights. SQL was used to extract and calculate the main metrics, while Excel was used to organize, group, and simplify the results for clearer interpretation.

To make the analysis more meaningful, time-based variables were grouped into broader categories:

- Months into seasons
- Days into weekday vs. weekend
- Hours into time blocks (morning, afternoon, evening, and night)

This approach made it easier to identify clear usage patterns and compare behavior between casual riders and annual members.

## Overview of Key Metrics

The analysis began with a general overview of the dataset to understand the overall distribution of rides and the main differences between user types.

Key findings include:

- Annual members account for the majority of total rides (64%), while casual riders represent 36%
- Electric bikes are the most commonly used bike type for both groups (approximately 65%), with only minor differences between user types
- Average ride duration, after excluding outliers, is:
    - 17.28 minutes for casual riders
    - 11.46 minutes for members

These findings show that casual riders generally spend more time per ride, while members tend to take shorter trips.

## Time-of-Day Patterns

To simplify interpretation, hours were grouped into four time blocks: morning, afternoon, evening, and night.

**Casual Riders**

- Morning: 16.00 minutes
- Afternoon: 19.65 minutes
- Evening: 16.06 minutes
- Night: 13.50 minutes

**Members**

- Morning: 10.53 minutes
- Afternoon: 11.75 minutes

- Evening: 11.94 minutes
- Night: 10.89 minutes

Casual riders record their longest rides in the afternoon, which aligns with leisure and recreational usage. Members show relatively stable ride durations across all time periods, indicating more consistent and routine behavior.

## Weekly Patterns (Day-Level Analysis)

To better understand behavioral differences, ride activity was analyzed across the days of the week.

The results show that:

- Members record higher ride activity during weekdays, indicating more regular and routine usage
- Casual riders show relatively higher activity during weekends, suggesting more flexible and recreational usage

Duration by Day Type

Ride duration was also compared between weekdays and weekends:

- Casual riders
    - Weekend: 19.49 minutes
    - Weekday: 15.97 minutes
- Members
    - Weekend: 12.48 minutes
    - Weekday: 11.14 minutes

This shows that casual riders not only ride more during weekends, but also spend more time per ride. Members, in contrast, maintain relatively stable ride durations throughout the week.

## Seasonal Trends

Monthly ride data was aggregated into seasons to highlight broader trends throughout the year.

The analysis shows that:

- Casual riders are highly influenced by seasonality
    - **Summer** records the highest usage (approximately **47.7%**)
    - **Winter** records the lowest usage (approximately **4%**)
- Members show a more consistent level of usage across all seasons, with less variation across the year

This suggests that casual riders are more affected by weather and seasonal conditions, while members rely on the service more regularly year-round.

## Ride Duration Distribution

To further understand ride behavior, ride durations were grouped into ranges.

The results show that:

- Casual riders have a larger share of rides in the **longer duration categories** (**20–40+ minutes**)
- Members are more concentrated in the **short-duration range**, especially **0–10 minutes** (approximately **57.8%**)

This indicates that casual riders are more likely to take longer rides, while members are more likely to take shorter trips.

## Key Behavioral Insights

The analysis reveals a clear and consistent behavioral difference between the two user groups.

Casual riders show more leisure-oriented behavior, characterized by:

- Longer ride durations
- Higher weekend activity

- Strong seasonal dependence
- Peak ride duration during the afternoon

Members show more routine-based behavior, characterized by:

- Shorter and more consistent ride durations
- Higher weekday activity
- Stable patterns across time and seasons

Overall, the results suggest that members use the service in a more regular and practical way, while casual riders display behavior that is more flexible, longer in duration, and more influenced by leisure-related factors.

# Conclusion and Recommendations

## Conclusion

The analysis identified clear behavioral differences between casual riders and annual members.

Casual riders tend to use Cyclistic bikes for leisure, as shown by longer ride durations, higher weekend usage, and strong seasonal patterns.

In contrast, annual members demonstrate more consistent and routine usage, with shorter ride durations and higher weekday activity, indicating commuting behavior.

These insights highlight a strong opportunity to convert casual riders into annual members through targeted marketing strategies.

## Recommendations

1. Target Weekend and Leisure Riders

Cyclistic should focus marketing campaigns on casual riders during weekends, when their usage is highest.

This can include weekend promotions and targeted digital ads to encourage membership conversion.

2. Introduce Seasonal Campaigns

Since casual riders are more active during peak seasons, especially summer, Cyclistic should launch seasonal promotions and limited-time membership offers to maximize conversions.

3. Promote Membership for Daily Use

Marketing should emphasize the benefits of membership as a convenient and cost-effective solution for daily transportation, encouraging casual riders to shift from occasional use to regular usage.